

SHETH L.U.J AND SIR MV COLLEGE
PRACTICAL NO:- 05

AIM :- Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

CODE :-

```
# Step 1: Publicly known values
p = 23
g = 5

print("Publicly Known Values:")
print("Prime Number (p):", p)
print("Primitive Root (g):", g)

# Step 2: Private keys
rani_private = 6
t128_private = 15

print("\nRani's Private Key:", rani_private)
print("T128's Private Key:", t128_private)

# Step 3: Generate Public Keys
rani_public = pow(g, rani_private, p)
t128_public = pow(g, t128_private, p)

print("\nRani's Public Key:", rani_public)
print("T128's Public Key:", t128_public)

# Step 4: Generate Shared Secret Key
rani_shared_key = pow(t128_public, rani_private, p)
t128_shared_key = pow(rani_public, t128_private, p)

print("\nRani's Shared Secret Key:", rani_shared_key)
print("T128's Shared Secret Key:", t128_shared_key)

# Step 5: Verify Shared Key
if rani_shared_key == t128_shared_key:
    print("\nKey Exchange Successful!")
    print("Shared Secret Key:", rani_shared_key)
else:
    print("\nKey Exchange Failed!")
```

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```
>>> ===== RESTART: C:/Users/IT-04/Documents/rani_P5_cyber.py =====
Publicly known values:
Prime number (p): 23
Primitive root (g): 5

Alice's Private Key: 6
Bob's Private Key: 15

Alice's Public Key: 8
Bob's Public Key: 19

Alice's Shared Secret Key: 2
Bob's Shared Secret Key: 2

Key Exchange Successful!
Shared Secret Key: 2
>>> ===== RESTART: C:/Users/IT-04/Documents/rani_P5_cyber.py =====
Publicly Known Values:
Prime Number (p): 23
Primitive Root (g): 5

Rani's Private Key: 6
T128's Private Key: 15

Rani's Public Key: 8
T128's Public Key: 19

Rani's Shared Secret Key: 2
T128's Shared Secret Key: 2
|
Key Exchange Successful!
Shared Secret Key: 2
>>>
```

CLI CODE :-

Publicly known values

p = 23

g = 5

Private keys

rani_private = 6

t128_private = 15

print("==== Diffie-Hellman Key Exchange =====")

print("\nPublicly Known Values:")

print("Prime Number (p):", p)

print("Primitive Root (g):", g)

print("\nPrivate Keys:")

print("Rani's Private Key:", rani_private)

print("T128's Private Key:", t128_private)

Generate public keys

rani_public = pow(g, rani_private, p)

t128_public = pow(g, t128_private, p)

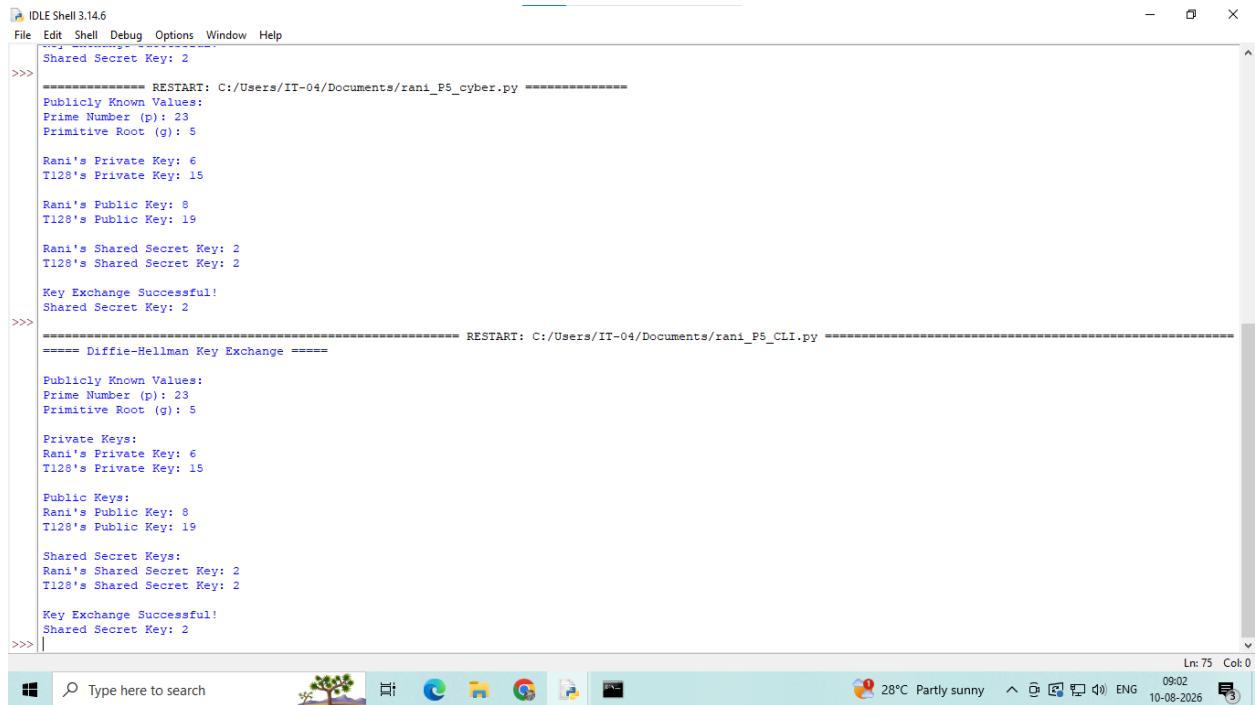
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```
print("\nPublic Keys:")
print("Rani's Public Key:", rani_public)
print("T128's Public Key:", t128_public)

# Generate shared secret keys
rani_shared_key = pow(t128_public, rani_private, p)
t128_shared_key = pow(rani_public, t128_private, p)

print("\nShared Secret Keys:")
print("Rani's Shared Secret Key:", rani_shared_key)
print("T128's Shared Secret Key:", t128_shared_key)

# Verify
if rani_shared_key == t128_shared_key:
    print("\nKey Exchange Successful!")
    print("Shared Secret Key:", rani_shared_key)
else:
    print("\nKey Exchange Failed!")
```



```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Shared Secret Key: 2
>>>
===== RESTART: C:/Users/IT-04/Documents/rani_P5_cyber.py =====
Publicly Known Values:
Prime Number (p): 23
Primitive Root (g): 5

Rani's Private Key: 6
T128's Private Key: 15

Rani's Public Key: 8
T128's Public Key: 19

Rani's Shared Secret Key: 2
T128's Shared Secret Key: 2

Key Exchange Successful!
Shared Secret Key: 2
>>>
===== RESTART: C:/Users/IT-04/Documents/rani_P5_CLI.py =====
===== Diffie-Hellman Key Exchange =====

Publicly Known Values:
Prime Number (p): 23
Primitive Root (g): 5

Private Keys:
Rani's Private Key: 6
T128's Private Key: 15

Public Keys:
Rani's Public Key: 8
T128's Public Key: 19

Shared Secret Keys:
Rani's Shared Secret Key: 2
T128's Shared Secret Key: 2

Key Exchange Successful!
Shared Secret Key: 2
>>>
```

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Diffie-Hellman Key Exchange - GUI

```
import tkinter as tk
from tkinter import messagebox

def perform_key_exchange():

    # Publicly known values
    p = 23
    g = 5

    # Private keys
    rani_private = 6
    t128_private = 15

    # Generate public keys
    rani_public = pow(g, rani_private, p)
    t128_public = pow(g, t128_private, p)

    # Generate shared secret keys
    rani_shared = pow(t128_public, rani_private, p)
    t128_shared = pow(rani_public, t128_private, p)

    # Update Rani values
    rani_public_value.config(text=str(rani_public))
    rani_shared_value.config(text=str(rani_shared))

    # Update T128 values
    t128_public_value.config(text=str(t128_public))
    t128_shared_value.config(text=str(t128_shared))

    # Update status
    if rani_shared == t128_shared:
        status_value.config(
            text="✓ KEY EXCHANGE SUCCESSFUL",
            fg="#00ff9d"
        )

    result_value.config(
        text="Shared Secret Key : " + str(rani_shared),
        fg="#00ff9d"
    )
```

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```
else:
    status_value.config(
        text="X KEY EXCHANGE FAILED",
        fg="#ff4d6d"
    )

    result_value.config(
        text="Keys do not match",
        fg="#ff4d6d"
    )

# ----- MAIN WINDOW -----

root = tk.Tk()

root.title("Diffie-Hellman Key Exchange")
root.geometry("850x650")
root.resizable(False, False)

# Background
root.configure(bg="#0b1020")

# ----- HEADER -----

header = tk.Frame(
    root,
    bg="#11182d",
    height=100
)
header.pack(fill="x")

title = tk.Label(
    header,
    text="🔒 DIFFIE-HELLMAN",
    font=("Segoe UI", 26, "bold"),
    bg="#11182d",
    fg="#00ff9d"
)
title.pack(pady=(18, 0))

subtitle = tk.Label(
```

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```
header,
text="Secure Key Exchange over an Insecure Network",
font=("Segoe UI", 11),
bg="#11182d",
fg="#9aa4bf"
)
subtitle.pack()

# ----- PUBLIC PARAMETERS -----

parameter_frame = tk.Frame(
    root,
    bg="#161d35"
)
parameter_frame.pack(pady=20, padx=40, fill="x")

parameter_title = tk.Label(
    parameter_frame,
    text="PUBLIC PARAMETERS",
    font=("Segoe UI", 10, "bold"),
    bg="#161d35",
    fg="#7f8cff"
)
parameter_title.pack(pady=(12, 5))

parameter_value = tk.Label(
    parameter_frame,
    text="Prime Number (p) = 23      |      Primitive Root (g) = 5",
    font=("Consolas", 13, "bold"),
    bg="#161d35",
    fg="ffffff"
)
parameter_value.pack(pady=(0, 12))

# ----- USERS AREA -----

users_frame = tk.Frame(
    root,
    bg="#0b1020"
)
users_frame.pack(padx=40, fill="x")
```

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----- RANI CARD -----

```
rani_card = tk.Frame(  
    users_frame,  
    bg="#151d34",  
    width=350,  
    height=240  
)  
rani_card.pack(side="left", padx=(0, 15), fill="both", expand=True)
```

```
rani_title = tk.Label(  
    rani_card,  
    text="👤 RANI",  
    font=("Segoe UI", 18, "bold"),  
    bg="#151d34",  
    fg="#ff6ec7"  
)  
rani_title.pack(pady=(18, 12))
```

```
rani_private = tk.Label(  
    rani_card,  
    text="Private Key : 6",  
    font=("Consolas", 12),  
    bg="#151d34",  
    fg="#d8def0"  
)  
rani_private.pack(pady=6)
```

```
rani_public_label = tk.Label(  
    rani_card,  
    text="Public Key :",  
    font=("Consolas", 12),  
    bg="#151d34",  
    fg="#d8def0"  
)  
rani_public_label.pack(pady=6)
```

```
rani_public_value = tk.Label(  
    rani_card,  
    text="—",  
    font=("Consolas", 14, "bold"),  
    bg="#151d34",  
    fg="#00d9ff"
```

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```
)
rani_public_value.pack()

rani_shared_label = tk.Label(
    rani_card,
    text="Shared Secret  :",
    font=("Consolas", 12),
    bg="#151d34",
    fg="#d8def0"
)
rani_shared_label.pack(pady=(12, 2))

rani_shared_value = tk.Label(
    rani_card,
    text="—",
    font=("Consolas", 18, "bold"),
    bg="#151d34",
    fg="#00ff9d"
)
rani_shared_value.pack()

# ----- T128 CARD -----

t128_card = tk.Frame(
    users_frame,
    bg="#151d34",
    width=350,
    height=240
)
t128_card.pack(side="right", padx=(15, 0), fill="both", expand=True)

t128_title = tk.Label(
    t128_card,
    text="🔒 T128",
    font=("Segoe UI", 18, "bold"),
    bg="#151d34",
    fg="#7f8cff"
)
t128_title.pack(pady=(18, 12))

t128_private = tk.Label(
    t128_card,
    text="Private Key   :   15",
```

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```
font=("Consolas", 12),
bg="#151d34",
fg="#d8def0"
)
t128_private.pack(pady=6)

t128_public_label = tk.Label(
    t128_card,
    text="Public Key    :",
    font=("Consolas", 12),
    bg="#151d34",
    fg="#d8def0"
)
t128_public_label.pack(pady=6)

t128_public_value = tk.Label(
    t128_card,
    text="—",
    font=("Consolas", 14, "bold"),
    bg="#151d34",
    fg="#00d9ff"
)
t128_public_value.pack()

t128_shared_label = tk.Label(
    t128_card,
    text="Shared Secret  :",
    font=("Consolas", 12),
    bg="#151d34",
    fg="#d8def0"
)
t128_shared_label.pack(pady=(12, 2))

t128_shared_value = tk.Label(
    t128_card,
    text="—",
    font=("Consolas", 18, "bold"),
    bg="#151d34",
    fg="#00ff9d"
)
t128_shared_value.pack()

# ----- EXCHANGE BUTTON -----
```

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```
exchange_button = tk.Button(  
    root,  
    text="⚡ PERFORM KEY EXCHANGE",  
    font=("Segoe UI", 13, "bold"),  
    bg="#00b894",  
    fg="white",  
    activebackground="#00d9a5",  
    activeforeground="white",  
    relief="flat",  
    cursor="hand2",  
    padx=30,  
    pady=12,  
    command=perform_key_exchange  
)  
exchange_button.pack(pady=22)
```

----- STATUS -----

```
status_value = tk.Label(  
    root,  
    text="● WAITING FOR KEY EXCHANGE",  
    font=("Segoe UI", 12, "bold"),  
    bg="#0b1020",  
    fg="#9aa4bf"  
)  
status_value.pack()
```

```
result_value = tk.Label(  
    root,  
    text="Shared Secret Key : —",  
    font=("Consolas", 14, "bold"),  
    bg="#0b1020",  
    fg="#ffffff"  
)  
result_value.pack(pady=10)
```

----- FOOTER -----

```
footer = tk.Label(  
    root,  
    text="RSA / DH Cryptography • Cybersecurity & Authentication",
```

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```
font=("Segoe UI", 9),  
bg="#0b1020",  
fg="#58627a"  
)  
footer.pack(side="bottom", pady=12)
```

----- RUN -----

```
root.mainloop()
```

